

SYSTEM 5.0: A Unified
Neuro-Semiotic Framework for
Symbolic Information Load, Limbic
Resonance, and Cultural Singularities
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Abstract

We propose SYSTEM 5.0, a comprehensive, physically grounded framework unifying neuroaesthetics, predictive coding, and cultural evolution. We formalize artistic value across three axes (Technique, Expressivity, Innovation) and four dimensions of verisimilitude (Biological, Spectral, Technical, Informational). Building on this, we introduce the theory of informational telepathy: the capacity of the human limbic system to predict intentions via the decoding of compressed symbolic information load. We model collective dynamics using symbolic frequency and limbic synchrony, explaining phenomena ranging from orchestral coordination to shared altered states of consciousness (e.g., dreams, near-death experiences, psychedelic visions) as instances of limbic resonance. Furthermore, we formalize the symbolic decoding gap to explain cultural shock in isolated populations. Finally, we demonstrate the framework's practical utility through proof-of-concept algorithmic attention engineering and discuss the implications of AI-generated super-stimuli. All mechanisms are falsifiable, with operational metrics provided.

Keywords: neuroaesthetics; informational telepathy; predictive coding; limbic resonance; symbolic decoding gap; altered states of consciousness; attention engineering

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Introduction

Formal theories of art, communication, and cultural evolution face a persistent tension between subjective phenomenological experience and measurable biological structure. Classical accounts often reduce art to resemblance or technical skill, while computational approaches attempt to quantify aesthetic value through image statistics or algorithmic complexity. Concurrently, human communication is frequently described metaphorically as "reading minds," yet no physical mechanism has been formally proposed to explain how static symbols enable one individual to predict the future actions of another with high accuracy without direct neural coupling.

SYSTEM 5.0 resolves these disparate threads into a unified neuro-semiotic framework. We propose that aesthetic evaluation, interpersonal prediction, and collective cultural states are all governed by the limbic system's capacity for predictive decoding of compressed symbolic information.

1.1 Core Hypotheses

H_1 (Multidimensional Art & Absurdity): Artistic value is a three-dimensional vector (Technique, Expressivity, Innovation) evaluated against four types of verisimilitude. "Perfection" is an operational threshold that triggers a measurable neurocognitive response of absurdity.

H₂ (Informational Telepathy): Symbols carry compressed informational load. High-load symbols activate hippocampal-amygdalar-prefrontal circuits, generating probabilistic behavioral models of the symbol's producer—a physical, biological form of telepathy.

H₃ (Cultural Shock & The Decoding Gap): Difficulties in cultural assimilation (e.g., isolated populations) are not cognitive deficits but predictive collapses caused by an exponentially widening symbolic decoding gap.

H₄ (Limbic Resonance & Altered States): Collective synchrony and shared altered states (dreams, NDEs, psychedelics) are manifestations of limbic resonance, occurring when external symbolic frequency drops, unmasking universal spectral archetypes.

H₅ (AI Super-Stimuli & Emotional Cryptography): AI acts as a hyper-accelerated super-stimulus producer, necessitating the evolution of "emotional cryptography" as a human defense mechanism against algorithmic limbic hijacking.

The Neuro-Semiotic Foundations of Art and Verisimilitude

2.1 The Three Axes of Artistic Value

Let W denote an artwork or symbolic artifact. We represent its artistic profile as a vector:

$$A(W) = (T(W), E(W), I(W)) \in [0, 10]^3$$

where T is Technique (mastery of medium), E is Expressivity (capacity to evoke limbic affect), and I is Innovation (departure from established conventions).

2.2 The Four Dimensions of Verisimilitude

Verisimilitude is decomposed into four components operating on distinct temporal scales:

Biological (VB): Similarity to human morphology (Timescale: 500–1,000 years).

Spectral (VS): Activation of archetypal, culturally stabilized visual patterns (Timescale: 50–100 years).

Technical (VT): Fidelity to physical reality via technological mediation (Timescale: 3–8 years).

Informational (VI): The density of compressed symbolic data activating limbic predictions (Timescale: 1–5 years).

2.3 Perfection and the Neurocognitive Response of Absurdity

An artwork W achieves operational perfection if:

$$\min\{T, E, I\} \geq 9.5 \wedge \min\{VB, VS, VT, VI\} \geq 0.95 \wedge Q(W) \geq 9.0$$

where $Q(W)$ is a composite quality score (effort, iteration, peer feedback, originality).

Perfection triggers Absurdity (A), a measurable response combining cognitive surprise and limbic reward:

$$A(W) = (\Delta NAcc(W), \tau_{fix}(W), R_{verb}(W))$$

where $\Delta NAcc$ is the relative increase in nucleus accumbens activation (fMRI), τ_{fix} is visual fixation duration (eye-tracking), and R_{verb} is the subjective report of awe or impossibility.

Informational Telepathy: Limbic Prediction via Symbolic Load

3.1 Informational Load

We define the informational load $L(s)$ of a symbol s as the total number of emotional-predictive bits activated in the limbic system:

$$L(s) = L_{semantic}(s) + L_{emotional}(s) + L_{predictive}(s) + L_{contextual}(s)$$

While an isolated letter carries 0–10 bits, a modern technical symbol (e.g., a legal contract or UI interface) carries 10^5 – 10^6 bits, requiring years of limbic calibration to decode.

3.2 The Four-Node Limbic Decoding Circuit

Informational telepathy operates via a specific neural circuit:

Hippocampus (Context Retrieval): Generates a contextual prior $P(\text{context} | s)$.

Amygdala (Valence Assignment): Assigns emotional valence $v(s) \in [-1, 1]$ and arousal $a(s) \in [0, 1]$.

Mirror Neuron System (Motor Simulation): Generates an embodied prediction of implied actions.

Prefrontal Cortex (Integration): Outputs a coherent predictive model $\hat{a}$ of the producer's next action a .

The formal prediction model is:

$$\hat{a} = \arg \max_a P(a | s, \text{context}, v(s), a(s))$$

Telepathic Accuracy is defined as $1 - H(\hat{a} | a)$, where H is the conditional entropy between predicted and actual actions.

Collective Dynamics: Frequency, Synchrony, and Resonance

4.1 Symbolic Frequency and Limbic Synchrony

Let f_{sym} be the symbolic frequency (rate of introduction of new symbols into an ecosystem).

Let $\Sigma \in [0, 1]$ be limbic synchrony (the degree to which a group decodes symbols with identical predictive models).

Highly synchronized groups (e.g., orchestras, elite dance troupes) exhibit Informational Telepathy in High Synchrony. The "sheet music" or "choreography" serves as a shared symbolic alphabet. Musicians do not merely react; they predict the actions of their peers via mirror-neuron simulation, achieving $\Sigma \rightarrow 1$.

4.2 Limbic Resonance in Altered States of Consciousness

SYSTEM 5.0 provides a non-mystical explanation for shared experiences in altered states (dreams, near-death experiences [NDEs], and psychedelic visions).

Under normal waking consciousness, high external f_{sym} dominates limbic processing. In altered states (e.g., via 5-HT_{2A} agonists like psilocybin, or hypoxia in NDEs), external $f_{\text{sym}} \rightarrow 0$. The cortical filter is inhibited, and the limbic system defaults to processing innate, high-load Spectral Archetypes (e.g., tunnels, light, geometric entities, parental figures). Because the human limbic hardware and spectral archetypes are universally shared, individuals in altered states experience Limbic Resonance—they report identical "realms" (e.g., heaven, hell, machine-elf encounters) not because these places exist externally, but because the limbic system is decompressing the same universal biological archive in the absence of external symbolic noise.

Cultural Evolution and Shock

5.1 The Symbolic Decoding Gap

Let SM be the modern symbolic ecosystem and SI be that of an isolated population. The symbolic decoding gap is:

$$\Delta_{\text{sym}} = L(SM) - L(SI \cap SM)$$

As technological complexity grows, $\Delta_{\text{sym}} \rightarrow \infty$. When an isolated individual encounters modern symbols, the hippocampus retrieves no context ($P \approx 0$), and the amygdala defaults to threat ($v < 0$, $a \rightarrow 1$). This results in Predictive Collapse—manifesting as cultural shock, anxiety, and withdrawal. This is a decoding failure, not a cognitive deficit.

5.2 Informational Colonialism

Because modern individuals share basal biological/spectral verisimilitude with isolated

populations, they can partially predict isolated behaviors. The reverse is impossible. This Asymmetric Telepathy maps directly onto colonial power structures, allowing dominant groups to manipulate or marginalize isolated groups via indecipherable symbolic systems (e.g., legal codes, bureaucratic interfaces).

Artificial Intelligence and Algorithmic Attention Engineering

6.1 AI as Super-Stimulus and Predictive Collapse

AI systems generate symbols with maximal $L(s)$ but lack genuine intentionality. This creates a prediction trap: the human limbic system generates models of AI "intentions" that never confirm in reality, leading to chronic predictive collapse, paranoia, and eventual rejection (necessitating Emotional Cryptography—the use of opaque, low-load symbols to evade AI decoding).

6.2 Proof-of-Concept: Limbic Hijacking

To prove that informational telepathy is exploitable for algorithmic amplification, we engineered two stress-test artifacts designed to maximally activate the 4-Node Limbic Circuit:

Text Artifact (280 characters): Utilizes autobiographical retrieval (Hippocampus), threat detection (Amygdala), and information gaps (PFC) to force high read-times and comment rates.

Video Artifact (15s AI-Generated): Combines uncanny valley micro-expressions, sudden audio silences, and pupillary dilation to trigger mirror-neuron simulation and amygdalar paranoia, yielding predicted completion rates $> 90\%$ and replay rates $> 20\%$.

Falsification Condition: If these artifacts fail to significantly outperform platform baseline engagement metrics, the theory of algorithmic limbic hijacking is weakened.

Operational Metrics and Falsifiability

Table 1. Proposed Experimental Measurements

Concept	Measurement Method	Falsification Criterion
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Informational Load $L(s)$	Free association latency + fMRI BOLD signal volume	
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No correlation between $L(s)$ and limbic activation volume		
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Telepathic Accuracy	Subject predicts action from text; compare to actual	
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Accuracy $\leq$ chance (25% for 4 options)		
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Symbolic Decoding Gap	Time to predict action from modern symbol (Isolated vs. Control)	
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Gap not significantly larger than within-group baseline		
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Limbic Resonance	fMRI hyperscanning during altered states / group rituals	
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Inter-subject correlation $\kappa \leq 0.2$		
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Absurdity $\Delta \text{NAcc (fMRI)} + \tau \text{fix (Eye-tracking)}$	No significant spike in reward/fixation metrics for	
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"perfect" art

Algorithmic Hijack A/B testing engagement metrics of engineered artifacts

No significant difference vs.

control posts

Discussion and Ethical Considerations

8.1 Relation to Existing Literature

8.2 Ethical Implications

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Informed Consent: Algorithmic attention engineering bypasses cortical deliberation, manipulating the amygdala without consent.

Vulnerable Populations: Isolated individuals and those with anxiety disorders are highly susceptible to predictive collapse and AI-induced paranoia.

Democratic Erosion: State or corporate actors could deploy high-L(s) artifacts to replace democratic deliberation with amygdalar reflex.

8.3 Limitations

Conclusion

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Conflicts of Interest

Data Availability

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Appendix A: Formal Notation Summary

Symbol Meaning

W Artwork / Symbolic Artifact

T, E, I Technique, Expressivity, Innovation scores

VB, VS, VT, VI Biological, Spectral, Technical, Informational Verisimilitude

$Q(W)$ Composite Quality score

$A(W)$ Absurdity response vector

s A symbol (letter, word, image, icon)

$L(s)$ Informational load of s (emotional-predictive bits)

$\hat{a}$, a Predicted action, Actual action

f_{sym} Symbolic frequency (rate of ecosystem update)

Σ Limbic synchrony (inter-subject correlation)

SM, SI Modern / Isolated symbolic ecosystems

Δ_{sym} Symbolic decoding gap

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